

National University of Computer and Emerging Sciences, Lahore Campus



Course: Electronic Circuit Design
Program: BS(Electrical Engineering)
Duration: 180 Minutes
Paper Date: 18-May-2017
Section: ALL
Exam: Final

Course Code: EE224
Semester: Spring 2017
Total Marks: 100
Weight: 50%
Page(s): 2
Questions: 5

- Q1 An amplifier for which $R_i = 1\text{ k}\Omega$, $R_o = 10\text{ k}\Omega$, and $A_o = 100\text{ A/A}$ is to be connected with a 100 mV source with a resistance of $100\text{ k}\Omega$ and a load of $1\text{ k}\Omega$. Identify the nature of amplifier model. What is the value of power gain (P_o/P_i), expressed in ratio and in dB. Draw the circuit schematic as well.

[20]
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$$i_o/i_i = 90.9$$

$R_o - \text{upar}$

- Q2 The ideal operational amplifier in the circuit shown in Figure 1 has $v_s = 9\text{ V}$. Find v_P , v_N , and v_O ?

[20]
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02

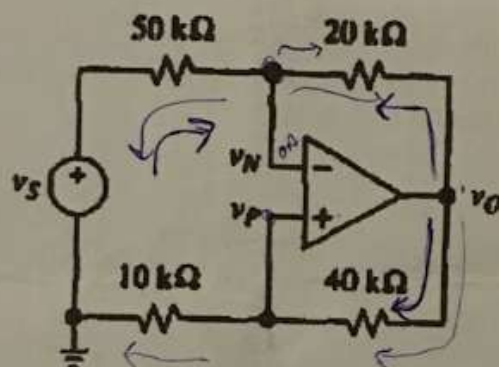


Figure 1

- Q3 The Figure 2 shows the circuit for an analog voltmeter of very high input resistance that uses an inexpensive moving-coil meter. The voltmeter measures the voltage V applied between the op amp's positive-input terminal and ground. Assuming that the moving coil produces full-scale deflection when the current passing through it is $100\text{ }\mu\text{A}$, design the value of R such that full-scale reading is obtained when V is $+10\text{ V}$.

[20]
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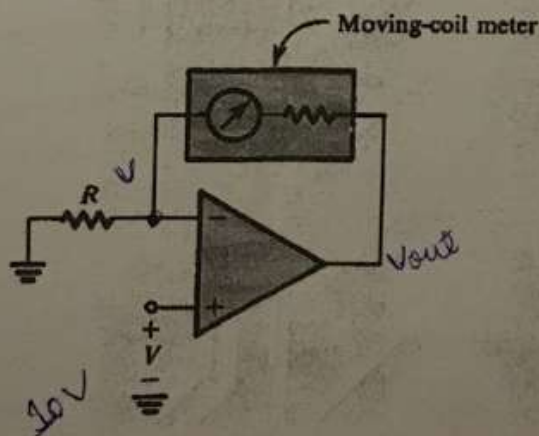
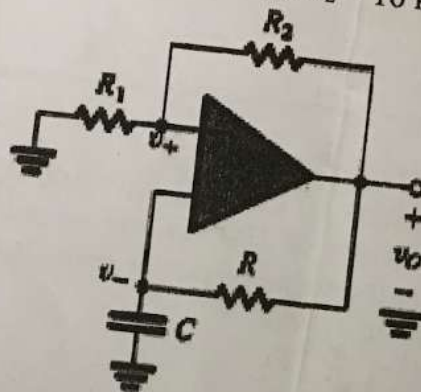


Figure 2

- Q4 (a) Draw the schematic of a monostable multivibrator using 555 timer circuit. Design the circuit for the value of external resistance (R), if an output pulse-width of 1.1-ms is to be generated using a $10\ \mu\text{F}$ capacitor. [20]
CLO 04
- (b) Identify the multivibrator circuit shown in Figure 3. Design the circuit for the value of R_1 , if the frequency of oscillation is 995 Hz, $R_2 = 16\ \text{k}\Omega$, $C = 10\ \text{nF}$ and $R = 62\ \text{k}\Omega$.



$$T = RC \ln(3)$$

Figure 3

- Q5 Draw the CMOS implementation of $Y = AB(C + D)$. Clearly show the synthesis process. Use any method for implementation. But **do NOT invert** the inputs (A , B , C and D) before applying them to the CMOS circuit. [20]
CLO 06